3.11 TORQUE ON A CURRENT LOOP

The force on a current carrying wire in a magnetic field is responsible for the motor operation.

3.11.1 Torque on a current loop placed in a magnetic field

Consider a rectangular loop PQRS carrying current I is placed in a uniform magnetic field $\vec{B}$. Let a and b be the length and breadth of rectangular loop respectively. The unit vector $\hat{n}$ normal to the plane of the loop makes an angle θ with the magnetic field as shown in Figure 3.57.

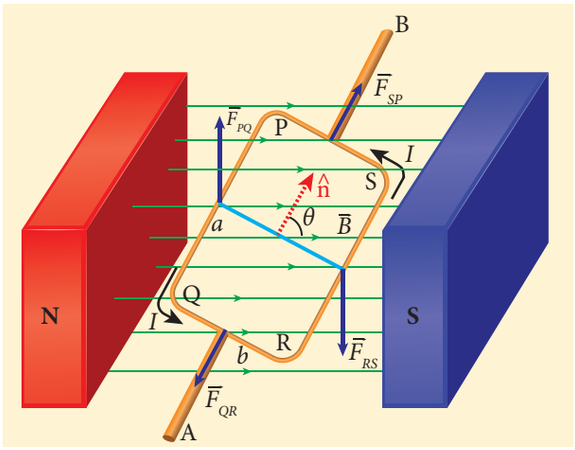


Figure 3.57 Rectangular coil placed in a magnetic field

The magnitude of the magnetic force acting on the current-carrying arm PQ is $F_{PQ} = IaB\sin\left(\pi/2\right) = IaB$. The direction of the force is found using right hand cork screw rule and its direction is upwards.

The magnitude of the force on the arm QR is $F_{QR} = IbB\sin\left(\pi/2 - \theta\right) = IbB\cos\theta$ and its direction is as shown in Figure 3.57.

The magnitude of the force on the arm RS is $F_{RS} = IaB\sin\left(\pi/2\right) = IaB$ and its direction is downwards.

The magnitude of the force acting on the arm SP is $F_{SP} = IbB\sin\left(\pi/2 + \theta\right) = IbB\cos\theta$ and its direction is also as shown in the Figure 3.57.

Since the forces F_{QR} and F_{SP} are equal, opposite and collinear, they cancel each other. But the forces F_{PQ} and F_{RS} , which are equal in magnitude and opposite in direction, are not acting along same straight line. Therefore, F_{PQ} and F_{RS} constitute a couple which exerts a torque on the loop.

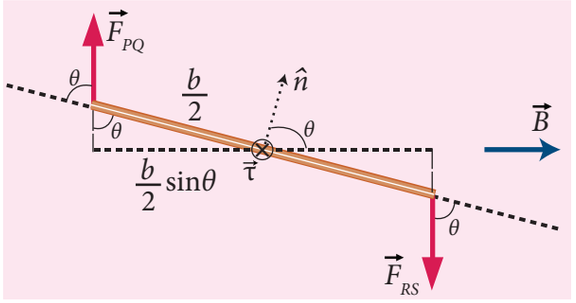


Figure 3.58 Side view of current loop

The magnitude of torque acting on the arm PQ about AB is $\tau_{PQ} = \left(\frac{b}{2}\sin\theta\right)IaB$ and it points in the direction of AB. The magnitude of the torque acting on the arm RS about AB is $\tau_{RS} = \left(\frac{b}{2}\sin\theta\right)IaB$ and points also in the same direction AB as shown in Figure 3.58.

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The total torque acting on the entire loop about an axis AB is given by

$$\tau = \left(\frac{b}{2}\sin\theta\right)F_{PQ} + \left(\frac{b}{2}\sin\theta\right)F_{RS}$$
$$= Ia(b\sin\theta)B$$

$\tau = IAB\sin\theta$ along the direction AB

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